# Competitive Exam Practice 3: Arrays and Stacks

GATE (Computer Science) Pattern

Auqib Hamid Lone

*Department of Computer Science & Engineering, GCET Ganderbal*

September 9, 2026

Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type. C uses 0-based indexing throughout.

**Q1** *[GATE CSE 1998 Q2.14, verified — GATEOverflow V2 p.199]* **[MCQ, 2 marks]** Let $A$ be a two dimensional array declared as follows:

```
A: array [1 ... 10] [1 ... 15] of integer;
```

Assuming that each integer takes one memory location, the array is stored in row-major order and the first element of the array is stored at location 100, what is the address of the element $A[i][j]$?

(A) $15i + j + 84$

(B) $15j + i + 84$

(C) $10i + j + 89$

(D) $10j + i + 89$

**Q2** *[GATE CSE 2005 Q5, verified — GATEOverflow V2 p.200]* **[MCQ, 1 mark]** A program $P$ reads in 500 integers in the range $[0, 100]$ representing the scores of 500 students. It then prints the frequency of each score above 50. What would be the best way for $P$ to store the frequencies?

(A) An array of 50 numbers

(B) An array of 100 numbers

(C) An array of 500 numbers

(D) A dynamically allocated array of 550 numbers

**Q3** *[Original, GATE-pattern]* **[NAT, 1 mark]** An integer array `A[100]` starts at base address 3000, and each `int` occupies 4 bytes. The address of `A[5]` is ______________. (Answer in integer)

**Q4** *[Original, GATE-pattern]* **[MCQ, 2 marks]** A $4 \times 5$ integer array `A` is stored in *column-major* order starting at base address 2000, with each element occupying 4 bytes. What is the address of `A[2][3]` (0-based indices)?

(A) 2052

(B) 2056

(C) 2044

(D) 2068

**Q5** *[GATE CSE 2004 Q3, verified — GATEOverflow V2 p.245]* [**MCQ, 2 marks**] A single array `A[1..MAXSIZE]` is used to implement two stacks. The two stacks grow from opposite ends of the array. Variables `top1` and `top2` (`top1` < `top2`) point to the location of the topmost element in each of the stacks. If the space is to be used efficiently, the condition for "stack full" is

(A) (`top1` = MAXSIZE/2) AND (`top2` = MAXSIZE/2+1)

(B) `top1` + `top2` = MAXSIZE

(C) (`top1` = MAXSIZE/2) OR (`top2` = MAXSIZE)

(D) `top1` = `top2` − 1

**Q6** *[GATE CSE 2015 Set 2 Q38, verified — GATEOverflow V2 p.246]* [**NAT, 1 mark**] Consider the C program below:

```
#include <stdio.h>
int *A, stkTop;
int stkFunc(int opcode, int val)
{
    static int size = 0, stkTop = 0;
    switch (opcode) {
        case -1: size = val; break;
        case 0: if (stkTop < size) A[stkTop++] = val; break;
        default: if (stkTop) return A[--stkTop];
    }
    return -1;
}
int main()
{
    int B[20]; A = B; stkTop = -1;
    stkFunc(-1, 10);
    stkFunc(0, 5);
    stkFunc(0, 10);
    printf("%d\n", stkFunc(1, 0) + stkFunc(1, 0));
}
```

The value printed by the above program is ____________. (Answer in integer)

**Q7** *[Original, GATE-pattern]* [**MCQ, 1 mark**] An integer stack of capacity 100 is implemented over an array with `top = -1` for the empty stack. Starting from empty, the operations `push(10)`, `push(20)`, `pop()`, `push(30)`, `push(40)`, `pop()` are performed in order. What are the final value of `top` and the value returned by `peek()`?

(A) `top` = 1, `peek()` returns 30

(B) `top` = 2, `peek()` returns 40

(C) `top` $= 1$, `peek()` returns 20

(D) `top` $= 0$, `peek()` returns 10

**Q8** *[GATE CSE 2004 Q5, verified — GATEOverflow V2 p.246]* [**MCQ, 1 mark**] The best data structure to check whether an arithmetic expression has balanced parentheses is a

(A) queue

(B) stack

(C) tree

(D) list

**Q9** *[GATE CSE 2014 Set 2 Q41, verified — GATEOverflow V2 p.246]* [**MCQ, 1 mark**] Suppose a stack implementation supports an instruction REVERSE, which reverses the order of elements on the stack, in addition to the PUSH and POP instructions. Which one of the following statements is TRUE (with respect to this modified stack)?

(A) A queue cannot be implemented using this stack.

(B) A queue can be implemented where ENQUEUE takes a single instruction and DEQUEUE takes a sequence of two instructions.

(C) A queue can be implemented where ENQUEUE takes a sequence of three instructions and DEQUEUE takes a single instruction.

(D) A queue can be implemented where both ENQUEUE and DEQUEUE take a single instruction each.

**Q10** *[GATE CSE 2015 Set 3 Q12, verified — GATEOverflow V2 p.246]* [**NAT, 2 marks**] The result of evaluating the postfix expression $10\ 5\ +\ 60\ 6\ /\ *\ 8\ -$ is ______________. (Answer in integer)